

SIMATS ENGINEERING
Slot: C, Course Code: ITA0501, Course Name: Computer Vision

Duration: 60 mins

Assessment Tool 1: Data Interpretation (High)

Marks: 25

Assessment Tool Weightage: 40%

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Q. No.	Questions																
1	Real-Time System Design Trade-off A surveillance system must operate under constraints: • Viola-Jones: fast but less accurate (78%) • YOLO: moderate speed, good accuracy (92%) • Deep Learning: high accuracy (95%) but slow Compare the techniques and evaluate which should be selected under strict latency constraints while maintaining acceptable accuracy. Justify your decision considering deployment conditions.																
2	Robust Detection under Environmental Variations Detection accuracy across environments: <table><tr><th>Method</th><th>Normal</th><th>Occlusion</th><th>Low Light</th></tr><tr><td>Viola-Jones</td><td>80%</td><td>60%</td><td>55%</td></tr><tr><td>YOLO</td><td>90%</td><td>75%</td><td>70%</td></tr><tr><td>Deep Learning</td><td>95%</td><td>85%</td><td>80%</td></tr></table> Compare the robustness of each method and evaluate which technique provides the best overall performance across varying conditions. Justify trade-offs.	Method	Normal	Occlusion	Low Light	Viola-Jones	80%	60%	55%	YOLO	90%	75%	70%	Deep Learning	95%	85%	80%
Method	Normal	Occlusion	Low Light														
Viola-Jones	80%	60%	55%														
YOLO	90%	75%	70%														
Deep Learning	95%	85%	80%														
3	Optical Flow Reliability vs Noise Sensitivity Two optical flow methods show: • Method A: stable vectors but misses small motions • Method B: detects fine motion but noisy vectors Compare both methods and evaluate which is more suitable for motion tracking in dynamic scenes. Justify based on application requirements.																
4	Background Modeling Strategy (GMM) A GMM-based system shows: • High false positives in dynamic background • Missed detections in static background Compare GMM performance across scenarios and evaluate whether GMM alone is sufficient. Propose and justify improvements.																
5	Motion Estimation vs Computational Cost Two methods: • Method A: low error (5%), high computation • Method B: moderate error (10%), low computation Compare the methods and evaluate their suitability for real-time vs offline applications. Justify your decision.																

Code of Conduct:

I SOMISETTY VENKATA SAI NAVEEN KUMAR (192425176) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student
SOMISETTY VENKATA SAI NAVEEN KUMAR
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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	MARKS
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

Signature of the Course Faculty
Dr P Lourdu Mahimai Doss

Answers

1. Real-Time System Design Trade-off

The three techniques have different speed–accuracy trade-offs:

Technique	Speed	Accuracy	Suitability
Viola-Jones	Very fast	78%	Good for strict real-time constraints
YOLO	Moderate/Fast	92%	Best balance
Deep Learning	Slow	95%	Best accuracy but high latency

Evaluation:

Viola-Jones provides the fastest response but its accuracy of 78% may be insufficient for a surveillance system. Deep Learning provides the highest accuracy (95%), but its slower processing makes it unsuitable when strict latency is required.

Decision: YOLO should be selected. It provides **92% accuracy with moderate speed**, giving a better balance between real-time performance and detection reliability. Although Deep Learning is slightly more accurate, the 3% improvement does not justify the additional computational delay under strict latency constraints.

Conclusion: YOLO is the most appropriate choice for a real-time surveillance system because it maintains high accuracy while satisfying real-time requirements.

2. Robust Detection under Environmental Variations

Given:

Method	Normal	Occlusion	Low Light	Average
Viola-Jones	80%	60%	55%	65%
YOLO	90%	75%	70%	78.33%
Deep Learning	95%	85%	80%	86.67%

Average accuracy:

Viola-Jones:

YOLO:

Deep Learning:

Evaluation:

Viola-Jones performs well under normal conditions but its accuracy decreases significantly with occlusion and low-light conditions. YOLO is more robust, with improved performance in both difficult conditions. Deep Learning achieves the highest accuracy in all three environments and has the smallest performance degradation.

Decision: Deep Learning is the most robust technique, with an overall average accuracy of **86.67%**.

Trade-off: Its major disadvantage is higher computational cost and potentially slower processing. If real-time speed is the highest priority, YOLO may be preferred. If robustness and accuracy are more important, Deep Learning is the better choice.

3. Optical Flow Reliability vs Noise Sensitivity

Method A: Stable vectors but misses small motions.

Method B: Detects fine motion but noisy vectors.

For a dynamic-scene motion-tracking application, both methods have advantages.

- **Method A** produces stable and reliable motion vectors and is less affected by noise. However, it may fail to detect small or subtle movements.
- **Method B** can detect fine movements, which is useful when tracking small objects or detailed motion. However, noisy vectors can produce false motion and reduce tracking stability.

Evaluation:

For a surveillance or general object-tracking system where stable tracking is important, **Method A is preferable**. For applications such as gesture recognition, micro-motion detection, or tracking very small movements, **Method B may be more suitable**.

Decision: For a **dynamic-scene motion-tracking system**, I would select **Method A** because stable vectors provide more reliable tracking and reduce false detections caused by noise.

Conclusion: Method A offers better reliability, while Method B offers greater sensitivity. The final selection depends on whether the application prioritizes **stability or detection of fine motion**.

4. Background Modeling Strategy — GMM

A Gaussian Mixture Model (GMM) represents each pixel using multiple Gaussian distributions. It can distinguish background and foreground by modeling different background states.

The given system has:

- High false positives in a dynamic background.
- Missed detections in a static background.

Problems:

1. **Dynamic background:** Moving trees, waving objects, shadows, and changing illumination can be incorrectly classified as foreground, producing false positives.
2. **Static background:** Objects that remain stationary for a long time may become incorporated into the background model, causing missed detections.

Evaluation:

GMM alone may not be sufficient for these challenging conditions. It requires improvements to adapt more effectively to environmental changes.

Recommended improvements:

- Use **adaptive learning rates** so the model can adjust to changing backgrounds.
- Use **shadow detection and removal** to reduce false foreground regions.
- Apply **morphological operations** to remove small noisy regions.
- Combine GMM with **object detection**, such as YOLO, for more reliable foreground identification.
- Use **temporal filtering** to distinguish actual moving objects from temporary background changes.

Conclusion: GMM is useful for basic background subtraction, but **GMM alone is insufficient** for highly dynamic scenes. Combining GMM with adaptive updating, shadow removal, morphological filtering, and object detection would significantly improve robustness.

5. Motion Estimation vs Computational Cost

Given:

Method	Error	Computational Cost
Method A	5%	High
Method B	10%	Low

Method A has lower error and therefore provides more accurate motion estimation. However, it requires high computational resources.

Method B has twice the error of Method A but requires much less computation.

Real-time application

For a **real-time application**, Method B may be preferable when processing speed and low latency are critical. A 10% error may be acceptable if the system must respond quickly.

Offline application

For an **offline application**, Method A is preferable because computational time is less critical. Its **5% error** provides significantly better motion-estimation accuracy.

Decision:

- **Real-time** → **Method B**, when speed and low computational cost are the priority.
- **Offline** → **Method A**, because accuracy is more important than processing time.

Conclusion: There is a direct trade-off between accuracy and computational cost. **Method A provides higher accuracy, whereas Method B provides faster and cheaper computation.** The appropriate method should therefore be selected according to the application requirements.

